



Universität
Zürich^{UZH}

Department of Informatics
Dynamic and Distributed Information Systems Group

Norms and Explanations for Recommender Systems

Lucien Heitz & Kathrin Wardatzky, March 11, 2026
Invited Talk @ ETH Zurich Social Networks Lab

1. Norms and News



2. Explanations and Evaluation



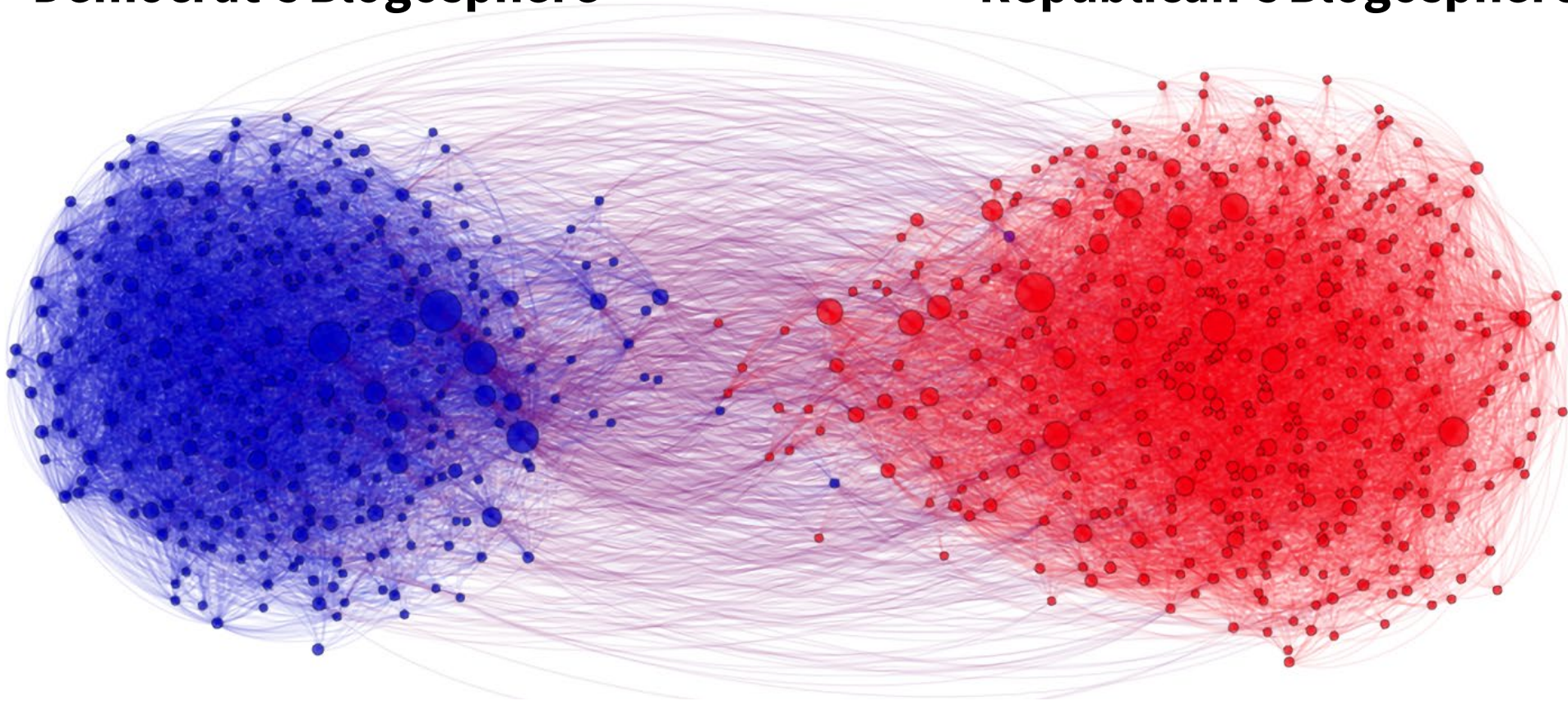
Part 1

Norms & News

Political Online News

Democrat's Blogosphere

Republican's Blogosphere



Visualizing Polarization in Political Blogs: <http://allthingsgraphed.com/2014/10/09/visualizing-political-polarization>

Research Agenda

**Goal: Create diverse news recommendations
to counteract political polarization.**

Step 1: Develop Diversity Algorithms

Step 2: Build Testing Infrastructure

Step 3: Conduct User Experiments

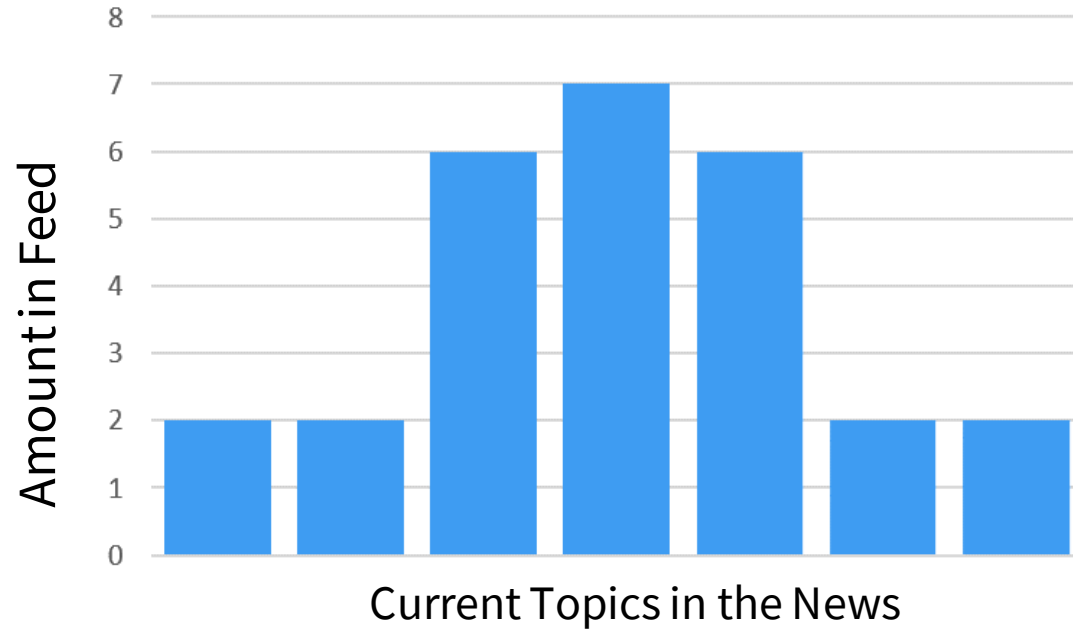
Step 1: Develop Diversity Algorithms

Based on:

Heitz et al., Informfully Recommenders – A Reproducibility Framework for Diversity-aware Intra-session Recommendations. Accepted at RecSys '25, 2025.

News Algorithms

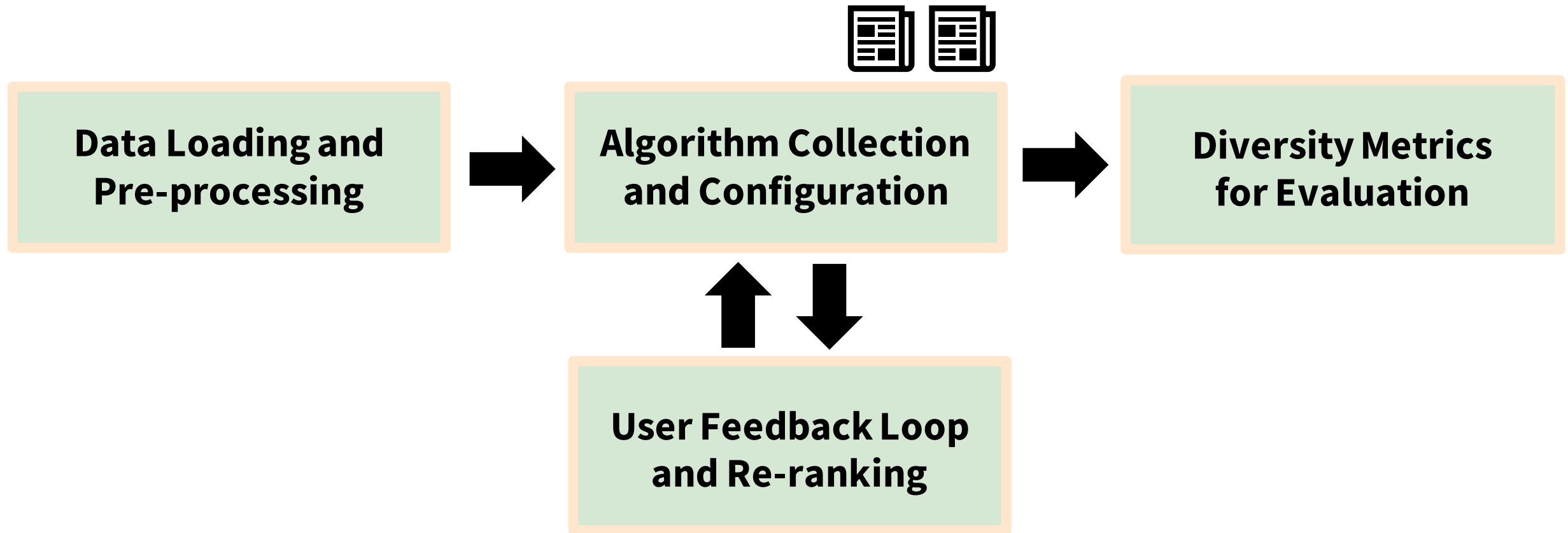
Participatory Diversity



Goal: Discuss common issues.

Model goals based on Helberger, N.: On the Democratic Role of News Recommenders. *Algorithms, Automation, and News*. Routledge, 2021.

Diversity Framework



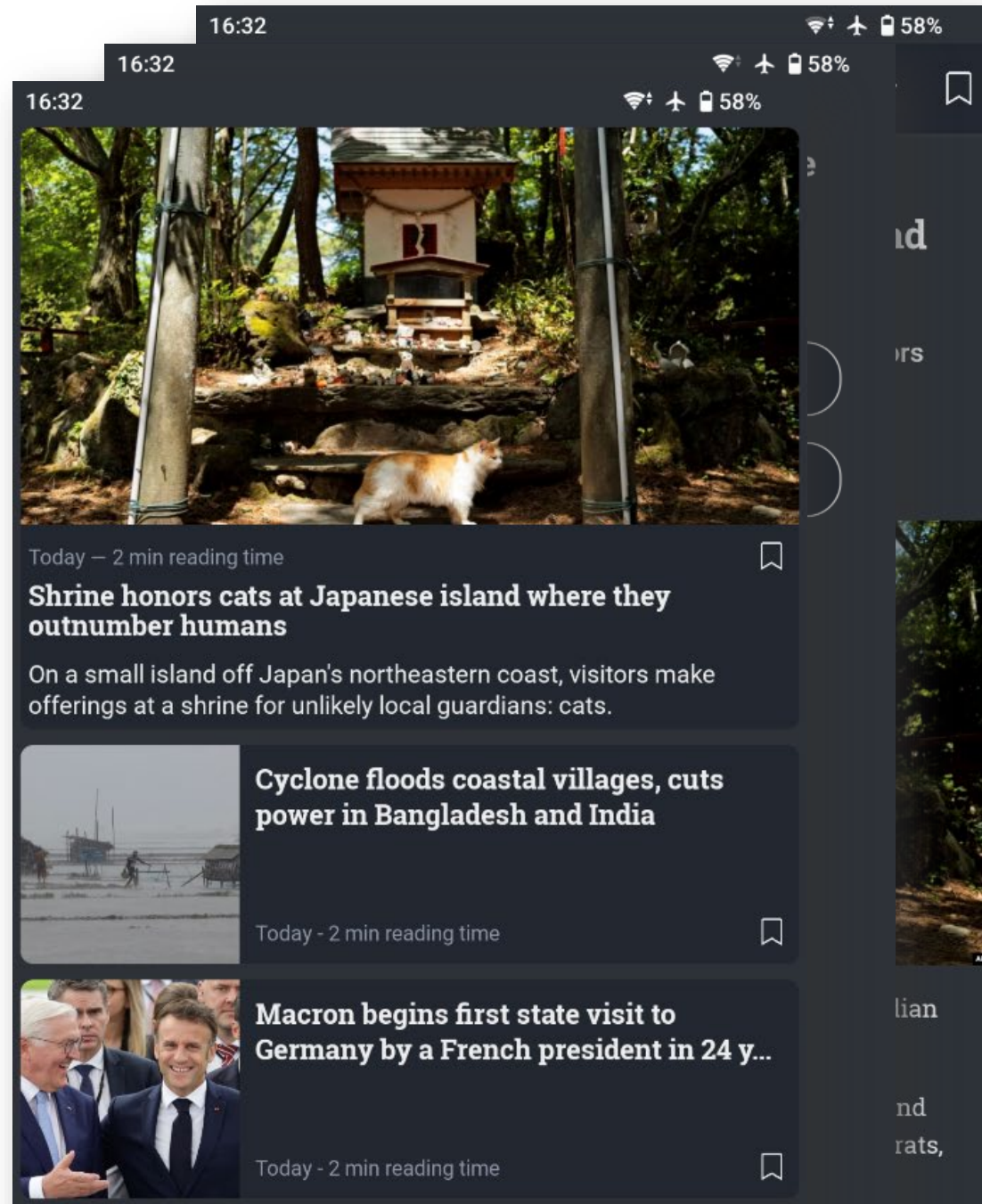
Step 2: Build Testing Infrastructure

Based on:

Heitz et al., Informfully – Research Platform for Reproducible User Studies. Published at RecSys '24, 2024.

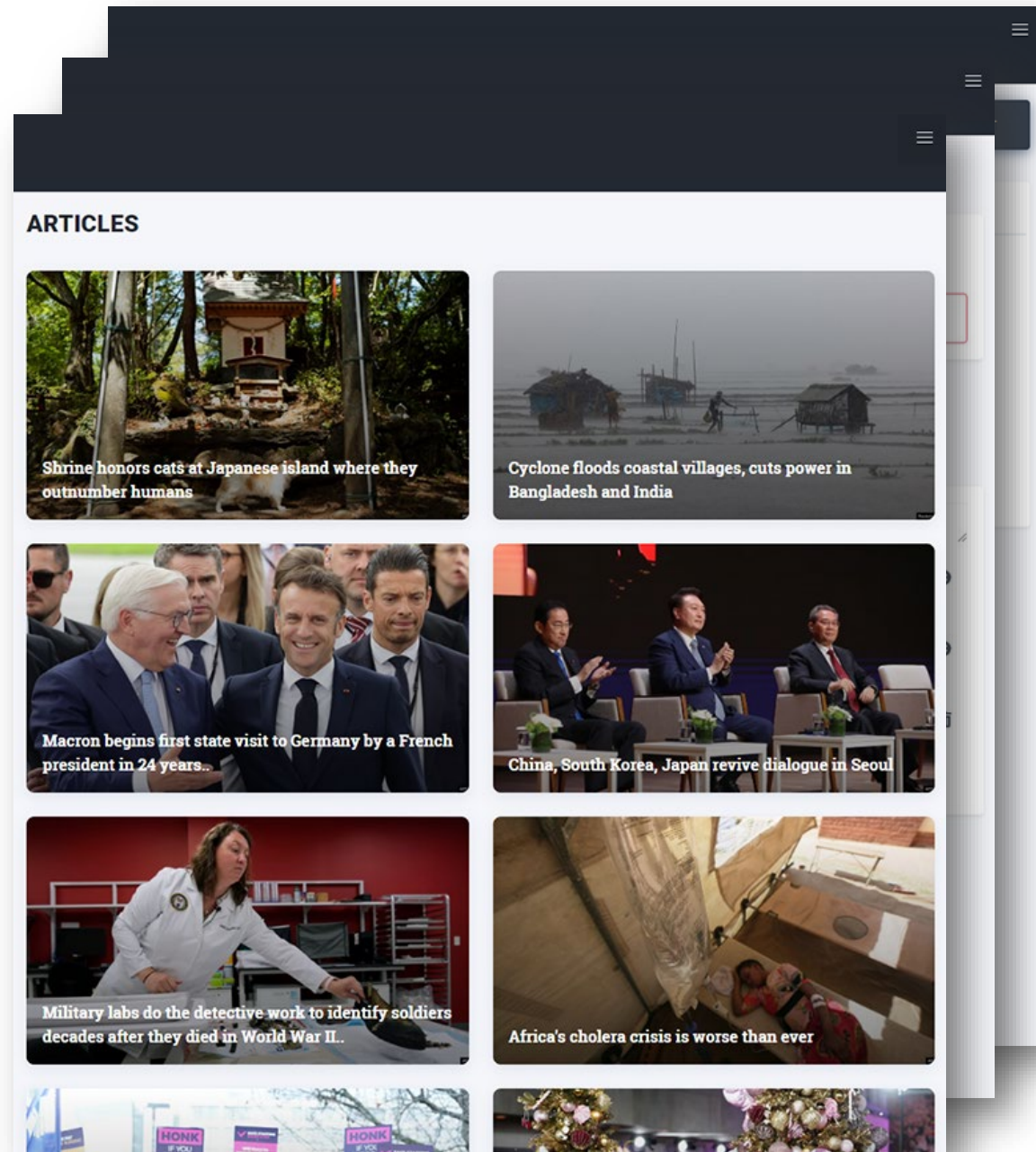
Mobile App

- Multimedia distribution platform
- In-app survey and tutorial section
- Android/iOS/Web



Admin Website

- Create user accounts
- Add and edit surveys
- Manage experiments



Step 3: Conduct User Experiments

Based on:

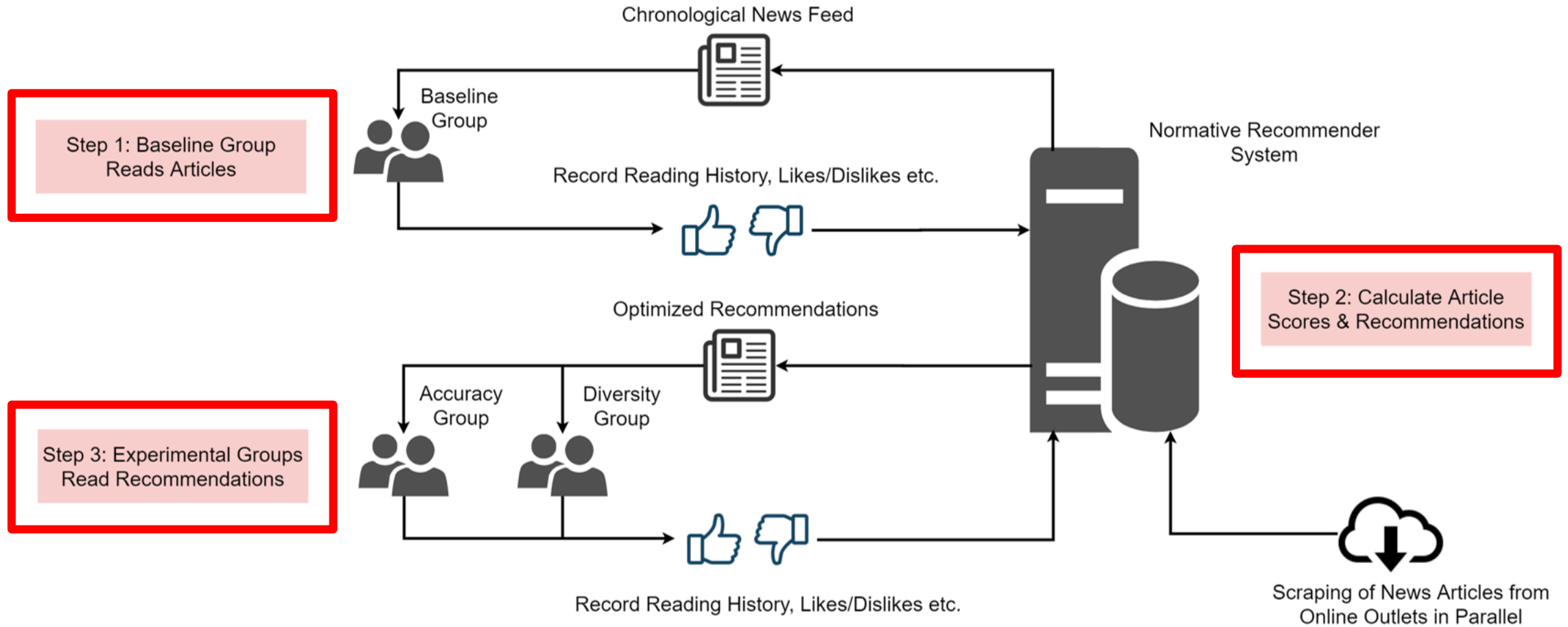
Heitz et al., Benefits of Diverse News Recommendations for Democracy: A User Study. Published in Digital Journalism 10(10), 2022.

Heitz et al., Deliberative Diversity for News Recommendations: Operationalization and Experimental User Study. Published at RecSys '23, 2023.

User Studies

	Switzerland	Germany
Participants	151	143
User Groups	3	2
Duration (Days)	35	28
Outlets	6	6
News Items	7,287	2,191
User Interactions	55k	17k

Experiment Setting



Selected Experimental Findings

CH	Baseline	Accuracy	Part. Div.
Reading Time (min)	400	398	406
Political Knowledge	3.4	3.6	3.2
Opposing Views Pref.	3.7	3.1	4.8
Appreciation of Journalism	3.2	4.1	4.3

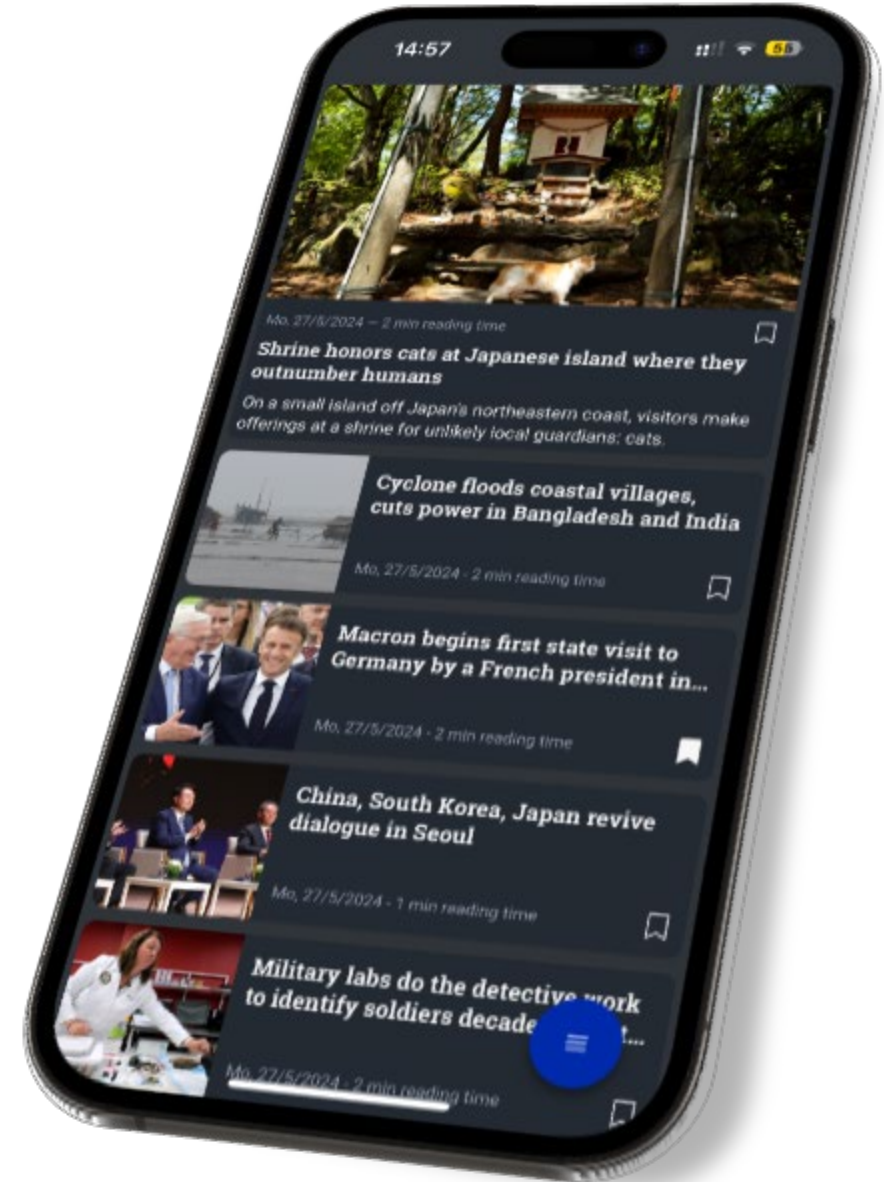
CH: Scores based on 7-point Liker scale self-assessment.

DE: Percentages based on yes/no questions and correct quiz answers (political knowledge).

Value significantly **higher** than the baseline. Value significantly **lower** than the baseline.

Outlook

- Get the app in people's hands
- Empirical studies and new models
- Maintain and support infrastructure



Part 2

Explanations & Evaluation

Introduction



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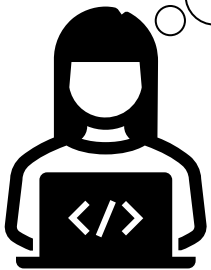


Recommendation

Explanation

Transparency

Which explanation is best
to evoke a certain effect
given **my** application
scenario?



Research Goal

Move towards a systematic explanation evaluation approach

Step 1: Understand the impact of user characteristics

Step 2: Understand the impact of explanation characteristics

Step 3: Build a pipeline for systematic evaluation

RQ1: How do user characteristics impact the explanation effects?

Based on:

Wardatzky et al., Whom do Explanations Serve? A Systematic Literature Survey of User Characteristics in Explainable Recommender Systems Evaluation. Published in ACM Transactions on Recommender Systems, 2025.

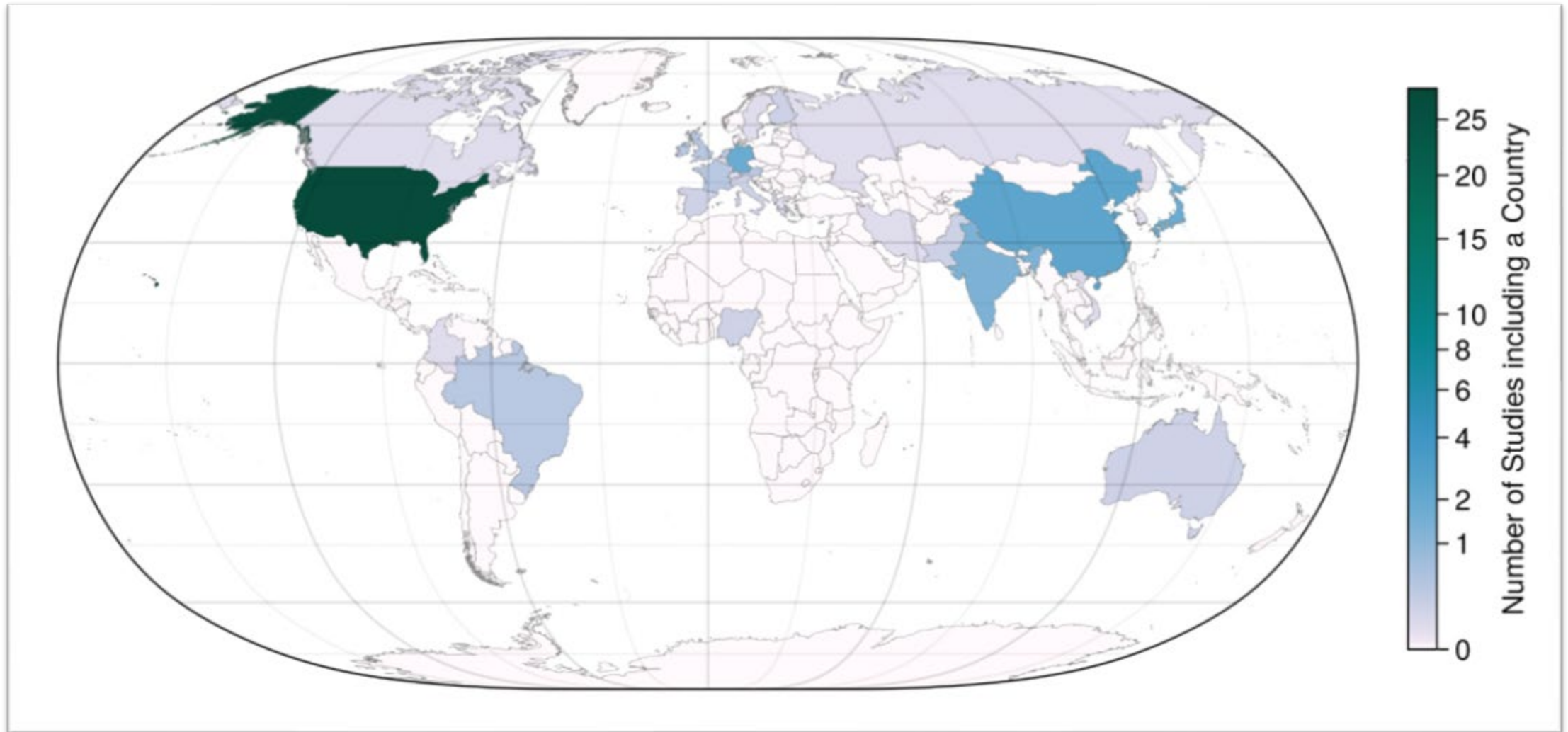
Survey Setup

- 124 papers
- RecSys explanations evaluated with user study

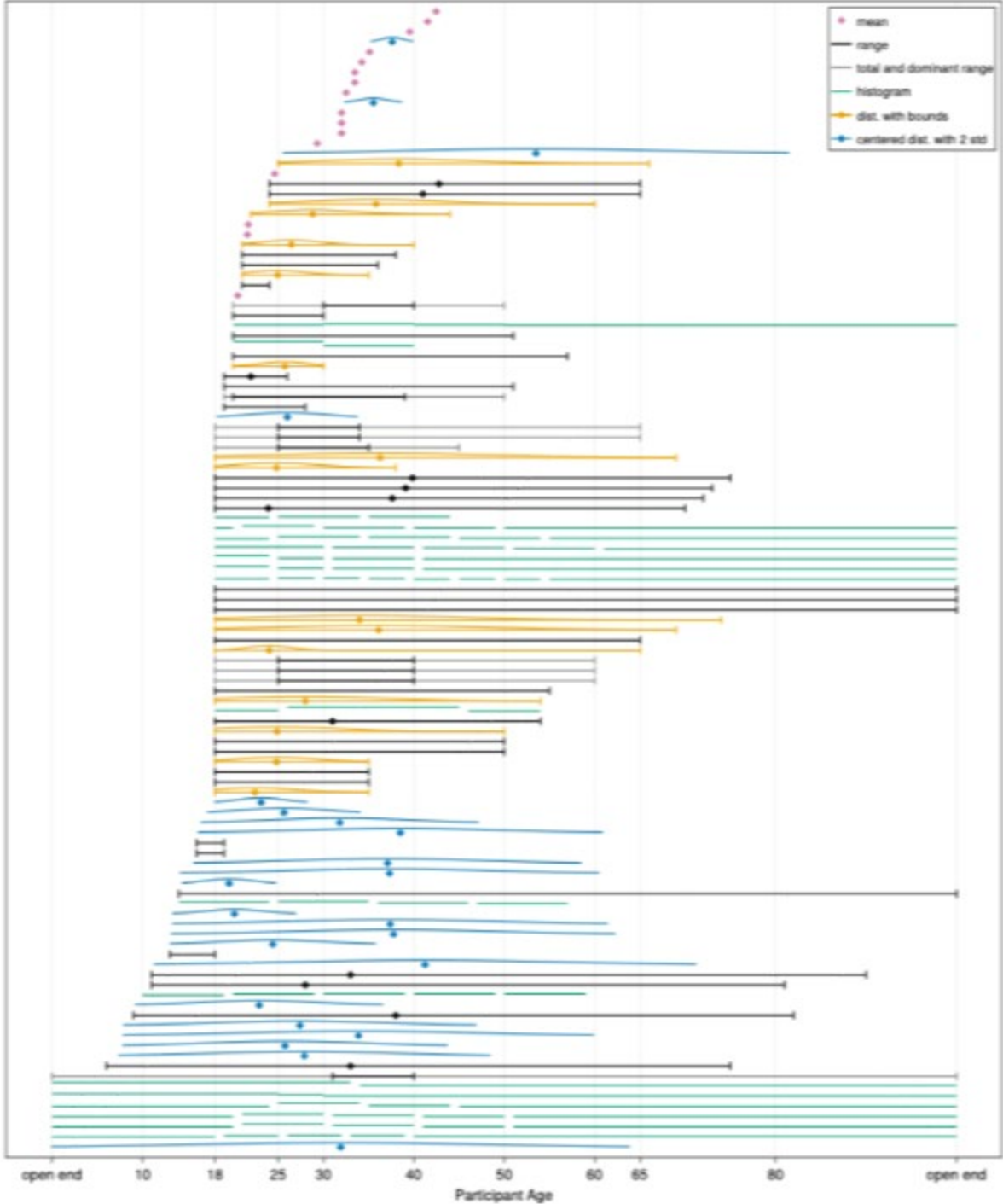
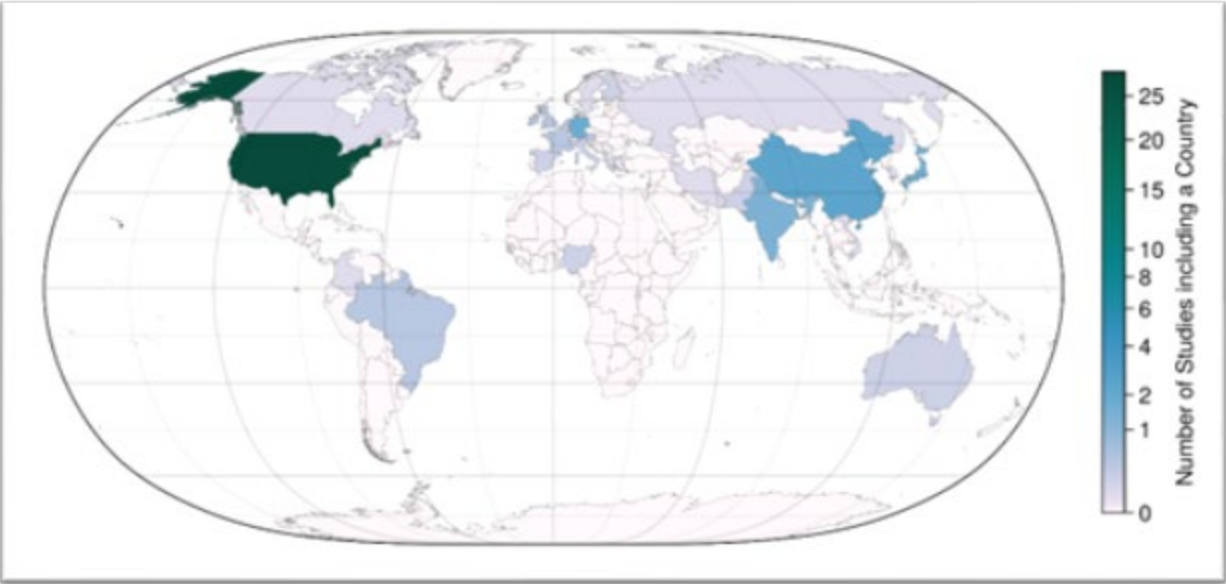
1. Participant characteristics in user studies
2. Effect of user characteristics



Participants are WEIRD



Participants are WEIRD



Characteristics Results

Domain Legend

- : document,
- : education,
- : health,
- : movie,
- : music,
- : online dating,
- : point-of-interest,
- : social.

User Characteristic	Effect Found	Effectiveness	Efficiency	Perceived Explanation Quality	Persuasiveness	Transparency
Demographic	Age	✓ ×		[149] 📄	[106] 📄	
	Location	✓ ×				[20] 📄
	Gender	✓ ×		[70] ❤️ [149] 📄, [34] 🕒	[106] 📄	[84] 🎓
	Education	✓ ×				[158] 🎓 [84, 158] 🎓
	Agreeableness	✓ ×		[100] 🕒 [101] 🕒	[2] 📄	
	Conscientiousness	✓ ×		[75] 📄	[2, 106] 📄 [2] 📄	
Personality	Decision-Making Type	✓ ×	[60, 61] 🕒 [60] 🕒	[34, 60, 61] 🕒 [34] 🕒		[61] 🕒 [60] 🕒
	Extraversion	✓ ×			[106] 📄 [101] 🕒	
	Need for Cognition	✓ ×		[88] 📄 [79] 📄, [92] 📄		[51] 📄 [26] 📄, [53] 📄
	Neuroticism	✓ ×			[106] 📄, [75] 📄 [2] 📄	
	Openness	✓ ×		[75] 📄	[2] 📄	
	Personal Innovativeness	✓ ×	[26] 📄 [53] 📄	[26] 📄 [53] 📄		[53] 📄 [26] 📄, [53] 📄
	Trust Propensity	✓ ×	[53] 📄	[53] 📄	[53] 📄	[53] 📄

RQ2: How do explanation characteristics impact the explanation effects?

Based on:

Wardatzky et al., The Impact of Explanation Characteristics on the Effects of Recommender Systems Explanations. Under Review at ACM Computing Surveys.

Survey Setup

- Extended to 180 papers
1. Measured effects
 2. Explanation characteristics
 - Modality
 - Information type
 3. Impact of characteristics on effects



Effects & Explanation Modalities

Displ.		Interact.	Result	Trust	#	Persuasiveness	#	Effectiveness	#	Satisfaction	#	Efficiency	#	Interface Qual.	#	Transparency	#	Other	#
hybrid	interactive	✓		[34, 53, 85]	8	[2]	1	[156]	1	[53, 61, 186]	5	[85]	2	[5, 10, 53, 61, 85]	16	[10, 53, 85]	4	[10]	17
		×		[10, 85, 141, 163]	16	[2, 141, 163]	16	[2, 141, 156, 157, 163, 188]	10	[61, 186]	4	[163]	1	[10, 61, 141, 188]	30	[10, 163, 188]	6	[10, 141, 188]	22
	static	✓		[23, 85, 145]	5			[180, 188]	8			[203]	1	[80]	1	[145]	1	[10]	4
		×		[23, 145, 203]	8			[31, 170, 180, 203]	10			[203]	1	[10, 170]	6	[23, 203]	2	[10]	6
textual	interactive	✓		[138, 172]	2			[77, 149, 193]	4	[138, 149, 165, 187]	8	[193, 198]	2	[18, 138, 149, 198]	8	[172]	3	[149]	1
		×		[81, 149, 172, 204, 213]	12	[18, 149]	2	[81]	2	[52]	1			[149, 154, 172]	7	[81, 149, 154]	24	[149]	2
	static	✓		[7, 15, 20, 21, 23, 24, 30, 35, 38, 46, 71, 94, 95, 101, 131, 132, 145, 153, 164]	90	[15, 20, 21, 35, 63, 70, 99, 112, 131, 132, 153, 161, 162, 164, 167, 191, 198]	94	[20, 21, 30, 35, 46, 71, 74, 86, 167, 169, 179, 197]	87	[7, 20, 94, 95, 131, 138, 153, 179, 185, 197, 198]	26	[35, 113, 153, 185, 198, 206]	38	[7, 10, 15, 35, 71, 94, 102, 112, 131, 132, 138, 153, 166, 198, 206]	54	[7, 20, 21, 24, 30, 35, 46, 71, 94, 112, 131, 132, 145, 164, 166, 179, 197, 208, 218]	64	[3, 10, 63, 95, 153, 185, 206]	62
		×		[10, 15, 21, 23, 24, 35, 38, 46, 71, 94, 101, 131, 132, 145, 153, 198]	108	[3, 15, 21, 35, 99, 131, 153, 161, 162, 167, 191]	68	[21, 30, 35, 46, 71, 86]	38	[52, 94, 131, 153, 169, 185, 197]	41	[35, 153, 185, 206]	37	[10, 15, 35, 71, 94, 101, 102, 131, 153, 166, 169, 198]	76	[10, 21, 23, 30, 35, 46, 71, 94, 102, 131, 166, 198, 201, 208]	84	[10, 153, 185, 206]	46
visual	interactive	✓		[53, 128]	3			[128]	1	[53]	1			[18, 91, 102, 107, 140]	7	[53]	1	[91, 107]	3
		×				[18]	1	[140]	1					[53, 102, 107, 140]	15	[102, 107, 111, 140]	5	[107]	1
	static	✓		[48, 153, 209]	5	[63, 153]	2	[8, 26, 180, 188, 209]	13	[8, 153, 209]	6	[153, 203, 209]	4	[26, 44, 62, 80, 153, 188, 209]	13	[8, 48, 182, 188]	11	[26, 63, 153, 188]	6
		×		[31, 48, 203]	8			[8, 26, 31, 180, 188, 203]	10	[8, 120]	4	[203]	4	[26, 31, 44, 188]	10	[8, 188, 201, 203]	11	[188]	2

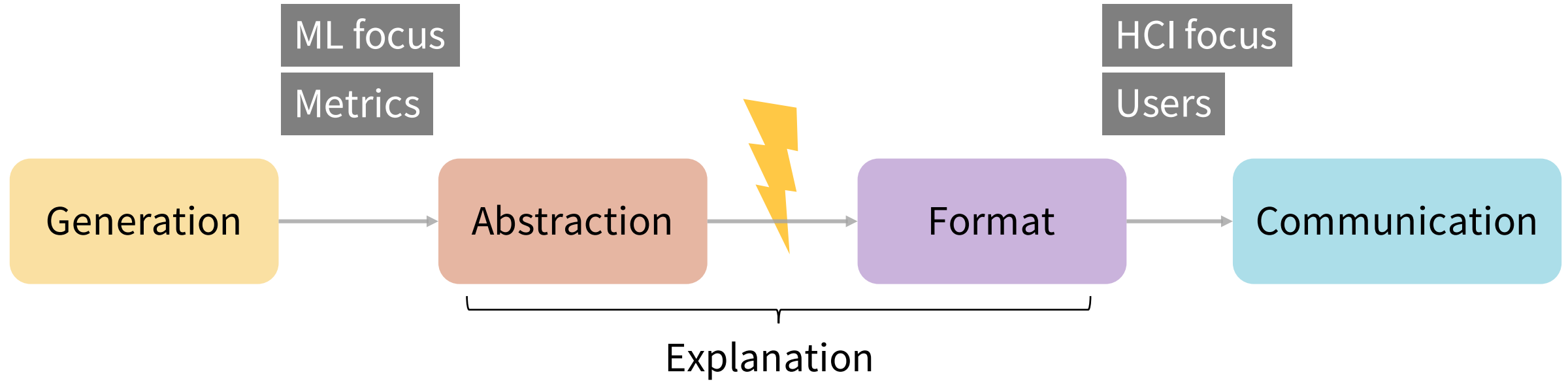
RQ3: How to move towards standardized evaluation of recommender systems explanations?

Based on:

Wardatzky et al., Toward Operationalizing a Comprehensive Evaluation Framework for Recommender Systems Explanations. Beyond Algorithms '25 co-located with RecSys, 2025.

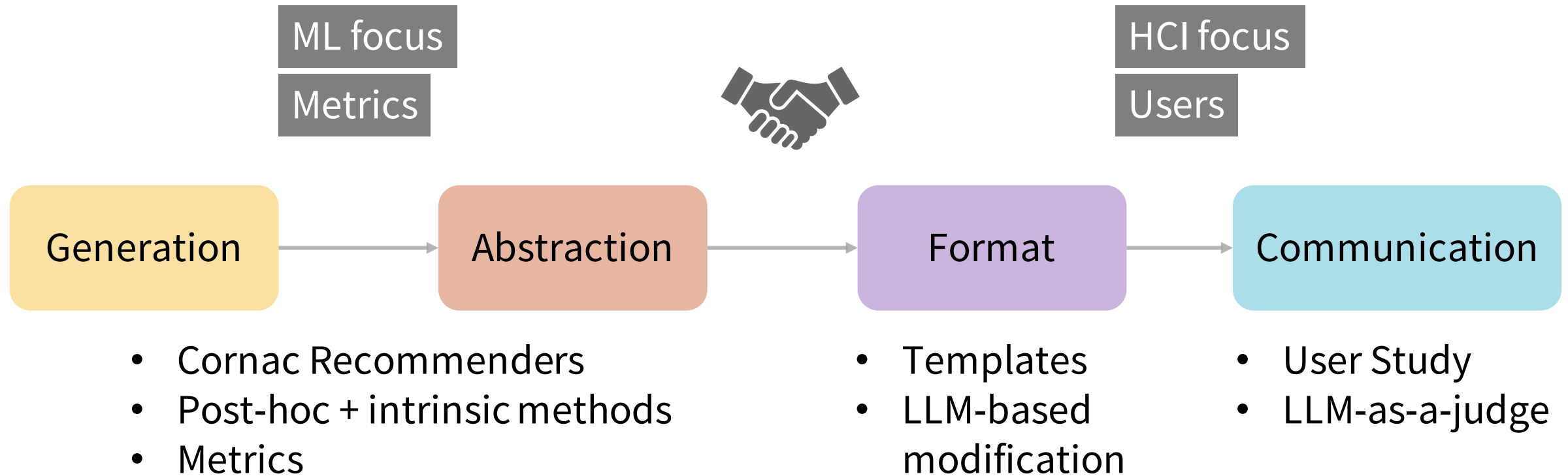
Work in Progress

Another lesson from the surveys



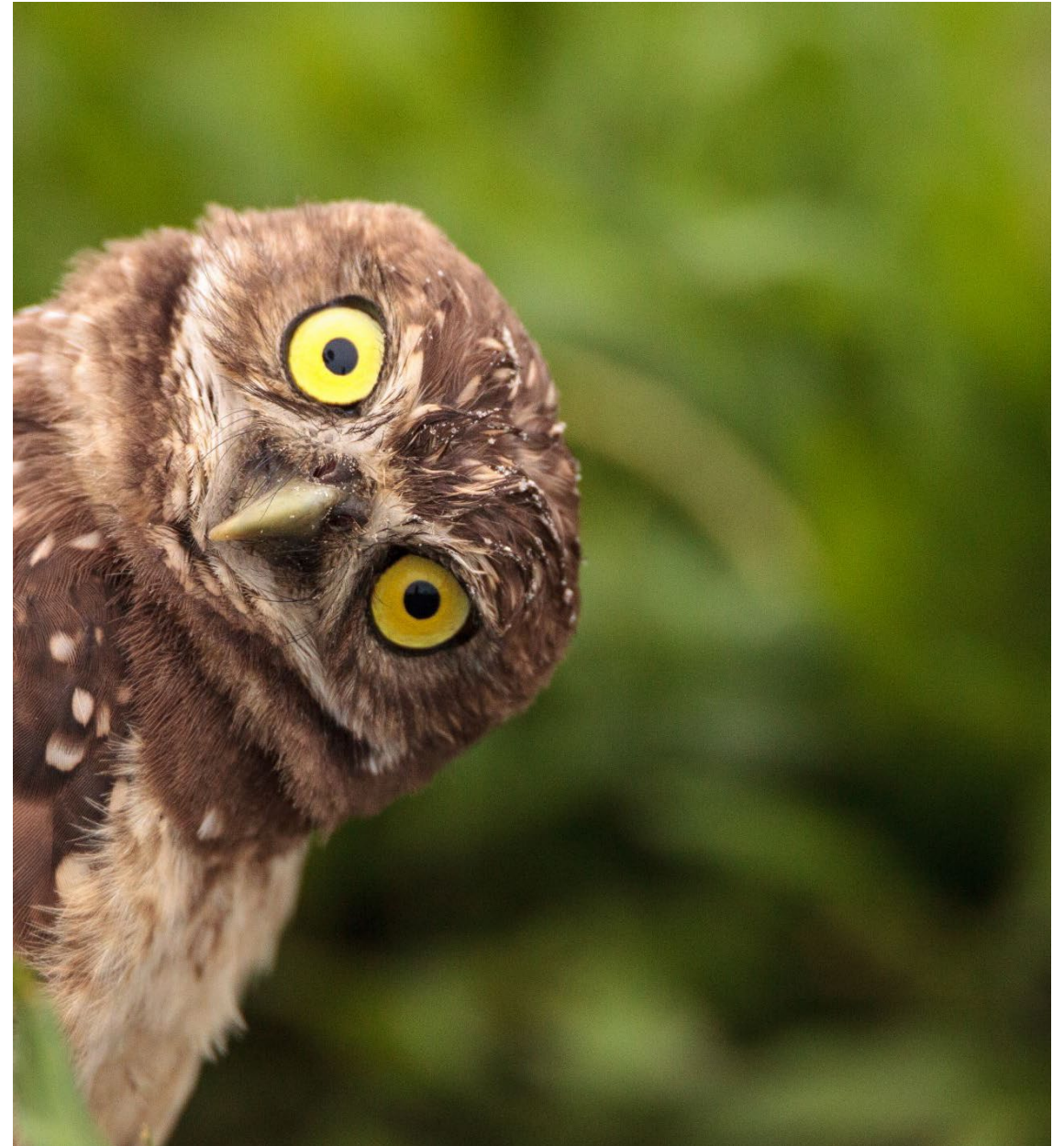
Donoso-Guzmán, et al. "Towards a comprehensive human-centred evaluation framework for explainable AI." *World Conference on Explainable Artificial Intelligence*. Cham: Springer Nature Switzerland, 2023.

The Framework



Summary

- Two literature surveys to investigate context:
 - Impact of user characteristics
 - Impact of explanation characteristics
- Evaluation framework for RecSys explanations (in progress)



Conclusion and Q&A

Thank you!

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NewsImages @ ICMR'26 - multimediaeval.github.io



Title: Bilston school takes a trip down memory lane as staff and pupils celebrate 90th anniversary.